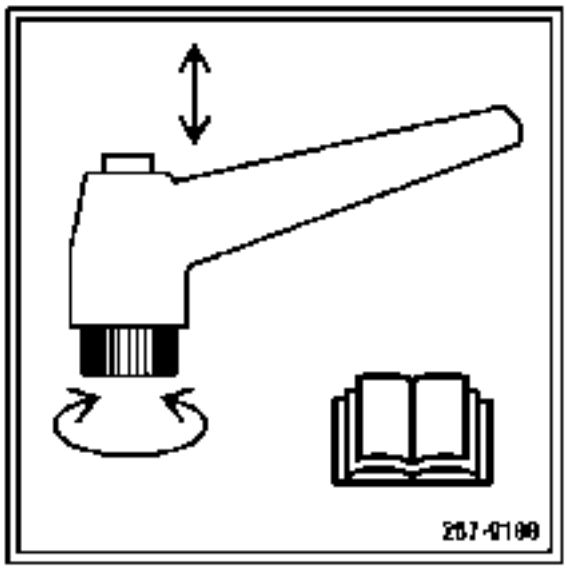


Illustration 52 g01114409

5. Tighten locking lever (3).

Note: In order to tighten the locking lever, perform the following procedure:

a. Turn the locking lever clockwise. Turn the locking lever by one-half turn.

b. Raise the locking lever.

c. Turn the locking lever counterclockwise by one-half turn.

d. Release the locking lever, and allow the locking lever to lower.

e. Repeat step 5a and step 5d until the locking lever is tight. Move the locking lever to the full counterclockwise position after the locking lever is tightened.

Lower

In order to lower the ROPS, perform the following steps for both folding joints on the ROPS:

1. Loosen locking lever (3).

Note: In order to loosen the locking lever, perform the following procedure:

a. Raise the locking lever.

b. Turn the locking lever clockwise by one-half turn.

c. Release the locking lever, and allow the locking lever to lower.

d. Turn the locking lever counterclockwise by one-half turn.

e. Repeat step 1a and 1d until the locking lever is loosened.

2. Remove the cotter pin and the locking pin in order to fold the ROPS toward the rear of the machine.

3. Lower the ROPS. Use two people to lower the ROPS.

4. If the machine is equipped with a canopy, perform steps 4a through 4d after lowering the ROPS.

a. Attach hold down cables (4).

b. Remove the canopy pins from position (5).

c. Raise the canopy.

d. Insert the pins into position (6).

5. In order to store locking pin (1), install locking pin (1) into the hole on the ROPS post.

6. Install cotter pin (2) through locking pin (1). The cotter pin will lock the locking pin in place.

i08286491

Drum Offset Control

SMCS Code: 6605

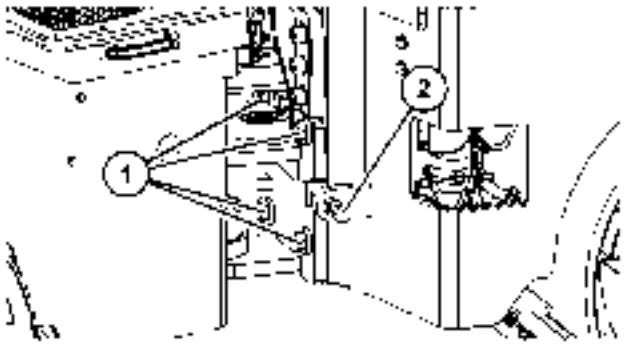


Illustration 53 g06207798

Left side of machine

(1) Slotted plate bolts
(2) Storage position

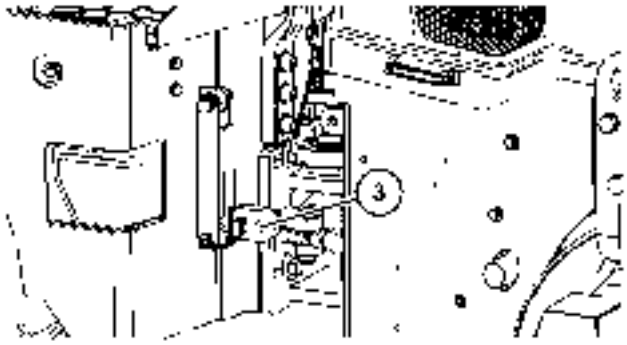


Illustration 54 g06207802

Right side of machine

(3) Threaded hole position

This machine can be shifted so that the rear drum is offset to the left side of the machine up to 40 mm (1.6 inch). Use this feature when necessary to compact close to a wall or other raised structure. To shift the drum, follow the steps below.

1. Make sure that the engine is off and the parking brake is on. This should be done on a smooth surface so the drum can slide easily when the machine is shifted. Then loosen four slotted plate bolts (1).

2. Loosen the jam nut until it is touching the head of the bolt.

3. Turn the bolt clockwise until the required offset is reached. The maximum offset is 40 mm (1.6 inch).

4. Tighten four slotted plate bolts (1).

5. Bolt is already in the storage position, tighten the jam nut so the bolt is retained.

To return the machine to normal alignment, insert the bolt in position (3) instead of position (2).

Engine Starting

i08579992

Engine Starting

SMCS Code: 1000; 7000

NOTICE
Do not crank the engine for more than 30 seconds. Allow the starter to cool for 2 minutes before cranking again.

Turbocharger damage can result, if the engine is not kept at low idle until the engine oil light/gauge verifies the oil pressure is sufficient.

The key must be in the RUN position with the engine on in order to maintain electrical functions, maintain hydraulic functions and prevent serious machine damage.

If the engine does not start, return the key to the OFF position before you return the key to START.

NOTICE
Do not use ether starting fluid in diesel engines equipped with inlet manifold heater.

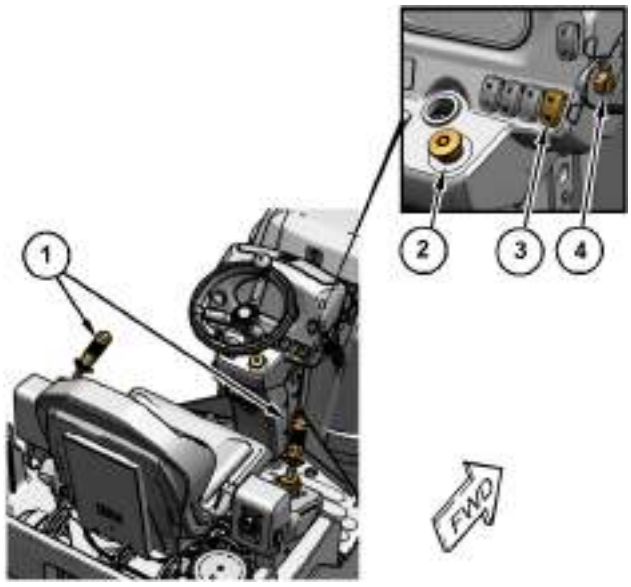


Illustration 55 g06612907

(1) Propel control levers
(2) Auxiliary shutdown knob
(3) Parking brake switch
(4) Key

Use the following procedure in order to start the engine.

1. Lift auxiliary shutdown knob (2).

2. Push down on the top side of parking brake switch (3) in order to apply the parking brake.

3. Move the propel control levers (1) to the NEUTRAL position.

4. Turn key (4) clockwise to the ON position to activate the glow plugs. Wait until the monitoring system indicates that the machine is ready to start.

Note: The glow plugs may not activate during higher ambient temperatures or higher engine temperatures.

5. Briefly sound the horn before you start the engine.

6. Turn key (4) clockwise to the START position. When the engine starts release the key.

Note: Engine will not start unless the park brake is engaged, the propel handle is in neutral.

i08287043

Engine and Machine Warm-Up

SMCS Code: 1000; 7000

NOTICE
Always run the engine at low idle for at least ten minutes before performing any other operations in cold conditions or each time the engine oil and oil filter are changed in order to protect your engine and hydraulic components.

NOTICE
Keep the engine speed slow until the indicator light for the engine oil pressure goes out.

If the light does not go out within ten seconds, stop the engine and investigate the cause before starting the engine again. Failure to correct the problem can cause engine damage.

NOTICE
Steering the machine completely to the steering stops while the hydraulic oil is below 0 °C (32 °F) can damage the steering system.

In temperatures below 0 °C (32 °F) work warm oil into the system by steering only a small amount at a time in each direction. Slowly increase the amount of each turn before steering completely to the steering stop.

This machine software controls Engine and Machine warmup by monitoring the hydraulic temperature. Until the hydraulic oil is up to a temperature suitable for normal use. The engine will be limited to low idle and the vibe system will not be enabled. The display will show a message alerting the operator that the message is in warmup mode. Once the message is no longer displayed, the machine may be operated normally and all machine functions will be available.

Parking

i08287074

Stopping the Machine

SMCS Code: 7000

NOTICE
Park the machine on a level surface. If it is necessary to park on a grade, securely block the tires and the drum.

Do not apply the parking brake while the machine is moving unless an emergency exists.

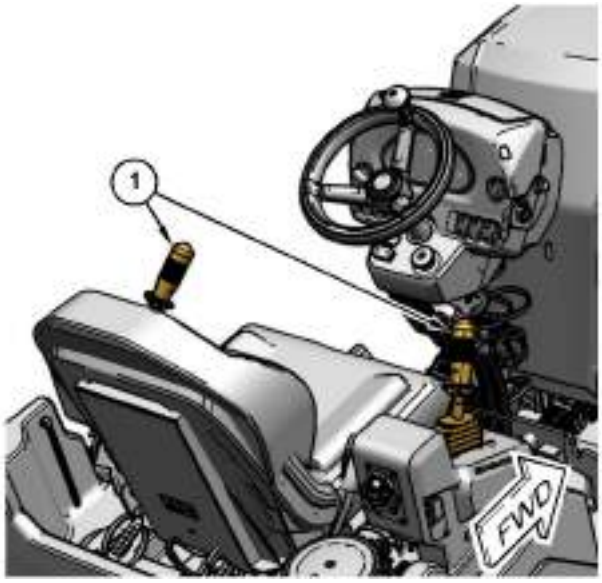


Illustration 56 g06612867

(1) Propel control levers



Illustration 57 g06612879

(2) Parking brake switch
(3) Throttle control switch

1. Move the propel control levers (1) to the NEUTRAL position.
2. Push low idle throttle control switch (3).
3. Push the top of the parking brake switch (2) to apply the parking brake.

Freezing Conditions

If freezing conditions are expected, the drums should be cleaned of mud and dirt. Park the machine on wood planks to prevent the drums from freezing to the ground.

Drain the water system and emulsion system (if equipped) to prevent damage.

i08287075

Stopping the Engine

SMCS Code: 1000; 7000

NOTICE
Stopping the engine immediately after it has been working under load, can result in overheating and accelerated wear of the engine components.

Follow the stopping procedure outlined below, in order to allow the engine to cool and prevent excessive temperatures in the turbocharger center housing, which could cause oil coking problems.

1. Before you stop the engine, allow the engine to run at low idle with no load for 5 minutes.

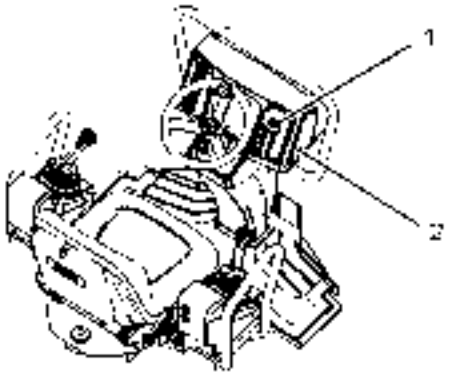


Illustration 58 g03179227

2. Push the top of the parking brake switch (1) in order to apply the parking brake.

3. Be sure that all the controls are in the OFF position.

4. Turn engine start switch (2) to the OFF position. Remove the key.

i02489691

Leaving the Machine

SMCS Code: 7000

1. Use the steps and the handholds in order to dismount the machine. Face the machine in order to dismount the machine.

2. Inspect the engine compartment for debris. Clean any debris from the engine compartment in order to avoid a fire. Clean any paper from the engine compartment in order to avoid a fire.

3. If the machine is being parked for an extended period of time, for example overnight, turn the battery disconnect switch to the OFF position.

4. Install all covers and locks in order to provide protection against vandalism.

i07735116

Machine Storage and Specified Storage Period

SMCS Code: 7000

Machine Storage

The Safety Section of this Operation and Maintenance Manual contains storage information for fuels, lubricants, and ether (if equipped).

The Operation Section of this Operation and Maintenance Manual contains information for short-term storage of this machine, including engine shutdown, parking, and instructions for leaving the machine.

For detailed steps on long-term storage refer to Special Instruction, SEHS9031, “Storage Procedure for Caterpillar Products”.

Specified Storage Period

The specified storage period of this machine is 1 year.

After the specified storage period has expired, consult your Cat dealer for inspect, repair, rebuild, install remanufactured, or install new components, and disposal options, and to establish a new specified storage period.

If a decision is made to remove the machine from service, refer to Decommissioning and Disposal for further information.

Transportation Information

i06972702

Shipping the Machine

SMCS Code: 7000; 7500

Investigate the travel route for overpass clearances. Make sure that there is adequate clearance for the machine that is being transported.

Remove ice, snow, or other slippery material from the loading dock and from the truck bed before you load the machine onto the transport machine. Removing ice, snow, or other slippery material will help to prevent the machine from slipping in transit.

NOTICE
Obey all state and local laws governing the mass, width and length of a load.

Observe all regulations governing wide loads.

When you move the machine to a colder climate, make sure that the cooling system has the proper antifreeze.

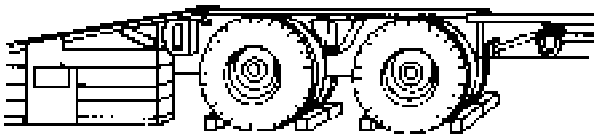


Illustration 59 g00303463

1. Before you load the machine, chock the trailer wheels or the rail car wheels, as shown.
2. Move the machine into position and engage the parking brake.

   If the machine is equipped with an optional canopy, then the machine should be facing rearward when the machine is positioned on the trailer.
3. Stop the engine and install the steering frame lock into the LOCKED position. The pin will hold the front frame and the rear frame rigid.

   Refer to the Operation and Maintenance Manual, “Steering Frame Lock” for further information.
4. Block the machine, and tie down the machine. Refer to the Operation and Maintenance Manual, “Lifting and Tying Down the Machine” for more information.
5. Lock the doors and the access covers. Attach any vandalism protection. Install the cover on the console. Cover the operator seat.
6. To protect the cooling systems, mix the solution of antifreeze and water. The solution should provide protection to the lowest expected outside temperature. Drain the excess coolant into a suitable container.
7. Perform a walk-around inspection and measure the fluid levels in the various compartments.
8. Travel at a moderate speed. Observe all speed limitations when you are roading the machine.

Consult your Caterpillar dealer for shipping instructions for your machine.

i08287851

Lifting and Tying Down the Machine

SMCS Code: 7000; 7500

⚠ WARNING

Improper lifting or tie-downs can allow load to shift and cause injury and damage.

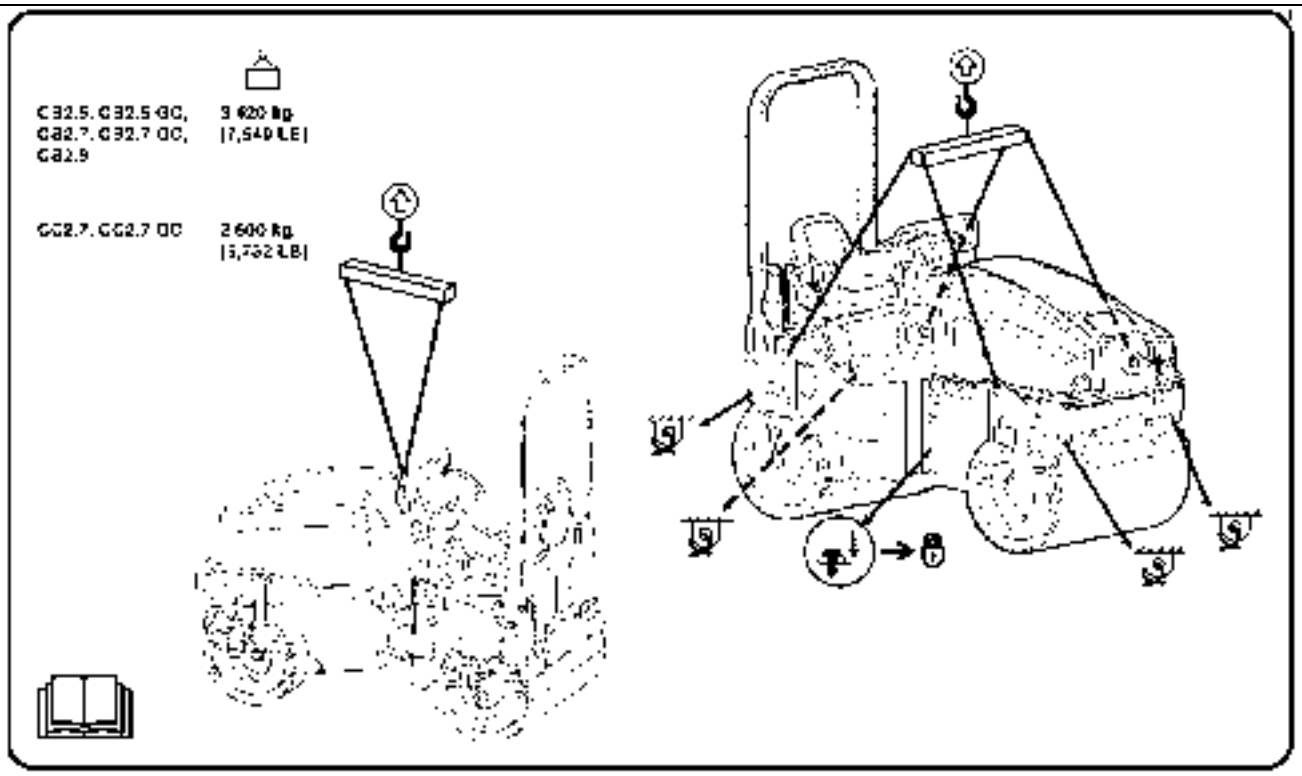


Illustration 60 g06612798

For the specifications of the machine, refer to Operation and Maintenance Manual, “Specifications”.

For specifications of any work tools, refer to the Operation and Maintenance Manual for the specific work tool.

Lifting Mass – The lifting mass indicates the maximum weight of the machine without any work tools.

Steering Frame Lock – The steering frame lock is used to secure the front frame and the rear frame from articulation.

Lock – Place all necessary locks in the locked position.

Lifting Point – To lift the machine, attach the lifting devices to the lifting points.

Tie Down Point – To tie down the machine, attach the tie-downs to the tie down points.

Lifting the Machine

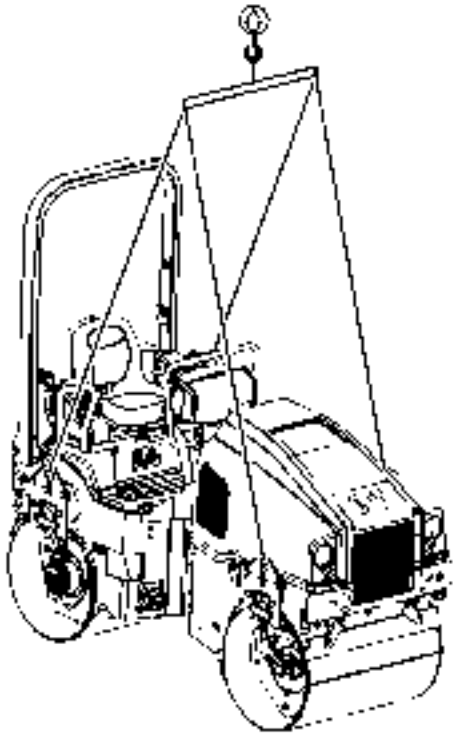


Illustration 61 g03166456

Multiple lifting points

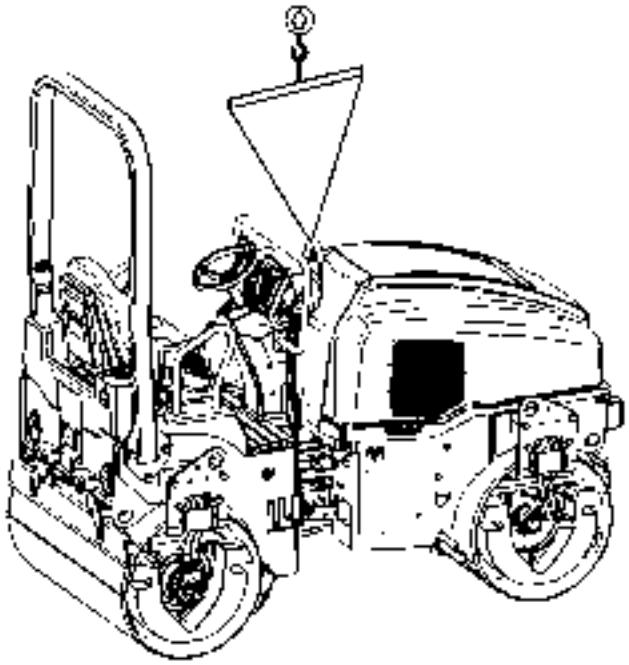


Illustration 62 g06184023

Single lifting point (if equipped)

Use properly rated cables and properly rated slings to lift the machine.

Position the crane or the lifting device to lift the machine in a level position.

1. Move the machine into position.

2. Apply the parking brake.

3. Turn the engine start switch key to the OFF position. Remove the key.

4. Install the steering frame lock pin.

5. Attach lifting cables to the lifting eyes. The lifting eyes are identified on the machine by labels. There are two lifting eyes on each side of the machine and one optional lifting eye on top of the machine.

6. Lift the machine slowly to make sure that the machine stays level. Move the machine to the desired position.

Tying Down the Machine

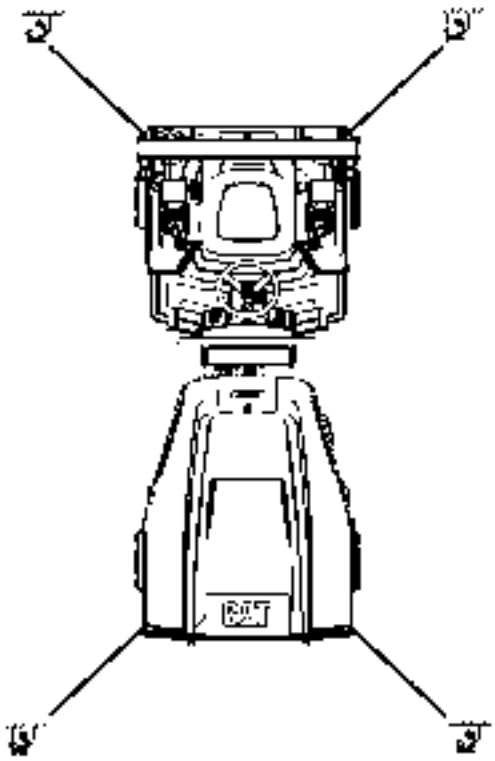


Illustration 63 g03166459

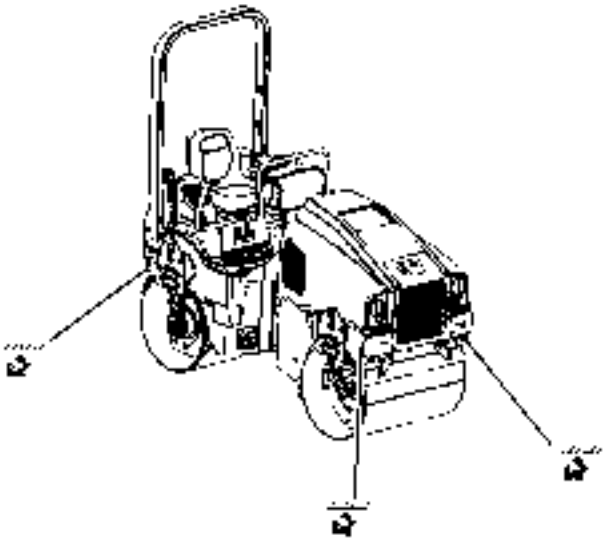


Illustration 64 g03166460

Do not tie down the machine over the operator platform. Tying down over the operator platform will reduce the life of the drum mounts and the isolation mounts for the platform.

Do not tie down the machine over the articulation hitch. Tying down over the articulation hitch will reduce the life of the articulation bearings, and could break the articulation hitch.

Use one chain per tie down point. Running a single chain through multiple tie down points can damage the scrapers, the spray bars, or other parts of the machine.

1. Move the machine into position.

2. Apply the parking brake.

3. Turn the engine start switch key to the OFF position. Remove the key.

4. Install the steering frame lock pin.

5. Place blocks under the front and the rear frames. The blocks will reduce the stress on the rubber isolation blocks.

6. Chock the front drum and the rear drum or tires.

7. Use the tie-down positions to secure the machine. The tie-down positions are identified on the machine by labels. There are two tie-down positions on each side of the machine.

Refer to Operation and Maintenance Manual, “Shipping the Machine” for shipping instructions.

Towing Information

i05006963

Towing the Machine

SMCS Code: 7000

WARNING

Improper hookup and towing is dangerous and could result in injury or death to yourself or others.

The towing connection must be rigid, or towing must be done by two machines of the same size as the towed machine. If two machines are used, connect a machine on each end of the towed machine.

If only one machine is used for towing, that machine must be larger than the towed machine.

Be sure that all necessary repairs and adjustments have been made before a machine that has been towed to a service area is put back into operation.

These towing instructions are for moving a disabled machine for a short distance at low speed. Move the machine at a speed of 2 km/h (1.2 mph) or less to a convenient location for repair. These instructions are only for emergencies. Always haul the machine if long distance moving is required.

Shielding must be provided on both machines. Shielding will protect the operator if the tow line or the tow bar breaks.

Do not allow an operator to be on the machine that is being towed unless the operator can control the steering and/or the braking.

Before towing, make sure that the tow line or the tow bar is in good condition. Make sure that the tow line or the tow bar has enough strength for the towing procedure that is involved. The strength of the towing line or of the tow bar should be at least 150 percent of the gross weight of the towing machine. This is true for a disabled machine that is stuck in the mud and for towing on a grade.

Keep the tow line angle to a minimum. Do not exceed a 30 degree angle from the straight ahead position.

Quick machine movement could overload the tow line or the tow bar. Overloading could cause the tow line or the tow bar to break. Gradual, steady machine movement will be more effective.

Normally, the towing machine should be as large as the disabled machine. Make sure that the towing machine has enough brake capacity, enough weight, and enough power. The towing machine must be able to control both machines for the grade that is involved and for the distance that is involved.

Sufficient control and sufficient braking must be provided when you are moving a disabled machine downhill. This may require a larger towing machine or additional machines that are connected to the rear. This will prevent the machine from rolling away out of control.

All situation requirements cannot be listed. Minimal towing machine capacity is required on smooth, level surfaces. On inclines in poor condition or on surfaces in poor condition, maximum towing machine capacity is required.

Attach the towing device and the machine before you release the brakes.

Consult your Caterpillar dealer for towing a disabled machine.

Running Engine

If the engine is running, the machine can be towed for a short distance under certain conditions. The power train and the steering system must be operable.

The operator must steer the machine that is towed in the direction of the tow line.

Ensure that all instructions in this section are followed carefully. Ensure that all instructions in this section are followed exactly.

Stopped Engine

WARNING

Shutting off the engine will result in the loss of machine steering.

When the engine is stopped, additional steps may be required before the machine is towed. In order to avoid damaging the power train, the steering system, and the brakes, which may be inoperable, additional steps may be required.

1. Block the drum or block the tires securely in order to prevent the movement of the machine. Do not remove the blocking until the tow vehicle has been positioned and the tow lines are in place.

NOTICE
Release the parking brake to prevent excessive wear and damage to the braking system when towing.

The procedure for manual release of the parking brake is outlined in the Operation and Maintenance Manual, “Parking Brake Manual Release”.

2. Manually release the parking brake. Refer to the Operation and Maintenance Manual, “Parking Brake Manual Release” for more information.

3. Attach the tow line to the machine at the tow points.

4. Attach the tow line to the vehicle that is used to tow the disabled machine.

5. Remove blocks from the drum or from the tires.

6. Tow the disabled vehicle at a slow rate of speed to the desired location.

7. Once the machine is at the desired location, securely block the drum or block the tires. Blocking the drums or tires will prevent movement of the machine.

8. Engage the parking brake.

9. Detach the tow lines.

i08518318

Parking Brake Manual Release

SMCS Code: 4267; 4354

⚠ WARNING

Personal injury or death can result from a brake malfunction. Do not operate the machine if the brake was applied due to a malfunction of the brake system.

Correct any problem before attempting to operate the machine.

Manual Brake Release With Hand Pump for CC2.7 GC

To release the parking brake manually on the CC2.7 GC, perform the following procedure:

1. To prevent the machine from moving, block the drum and block the tires securely.

2. To hold the front frame and the rear frame rigid, install the steering frame lock into the LOCKED position.

Note: Refer to “Steering Frame Lock“ for further information.

3. Open the engine compartment. Refer to “Access Doors and Covers“.

4. Ensure that the propel lever is in the NEUTRAL position. Refer to “Operator Controls“. If the propel lever is not in the NEUTRAL position, the hand pump will not build pressure.

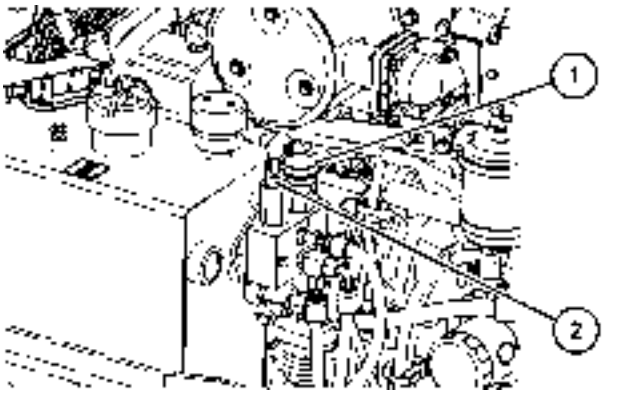


Illustration 65 g06695657

(1) Needle valve
(2) Hand pump

5. Close needle valve (1). The setscrew in the knob of needle valve (1) must be released to open needle valve (1).

6. Actuate hand pump (2) until pump (2) cannot be further pumped. Approximately five to seven full strokes of the pump will be required. The brakes are now released.

7. To engage the brakes, open needle valve (1).

Manual Brake Release for CB2.5 GC, and CB2.7 GC

To release the parking brake manually on the CB2.5 GC, and CB2.7 GC drum propel motors, perform the following procedure:

1. To prevent the machine from moving, block the drum securely.

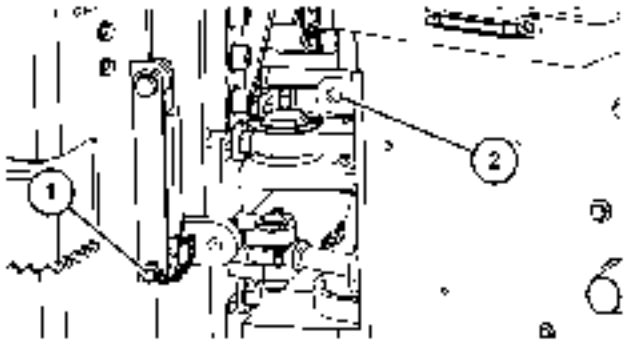


Illustration 66 g06183886

UNLOCKED position

(1) Pin
(2) Eye Hole

2. To hold the front frame and the rear frame rigid, install the steering frame lock into the LOCKED position.

Note: Refer to “Steering Frame Lock“ for further information.

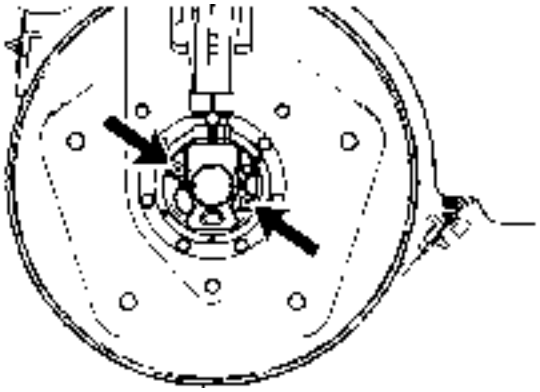


Illustration 67 g03179280
Front drum

3. Remove the plugs to access the release screws for the front drum propel motor which is on the left side of the drum.

4. Evenly tighten the two release screws to compress the brake springs by approximately two turns.

5. Repeat step 3 and step 4 to release the rear drum propel motor which is on the right side of the rear drum.

Note: Rotate the screws often to ensure that the brake does not become misaligned and not fully released.

Engine Starting (Alternate Methods)

i03650200

Engine Starting with Jump Start Cables

SMCS Code: 1000; 7000

⚠ WARNING

Failure to properly service the batteries may cause peronal injury.

Prevent sparks near the batteries. They could cause vapors to explode. Do not allow the jump start cable ends to contact each other or the machine.

Do not smoke when checking battery electrolyte levels.

Electrolyte is an acid and can cause personal injury if it contacts skin or eyes.

Always wear eye protection when starting a machine with jump start cables.

Improper jump start procedures can cause an explosion resulting in personal injury.

Always connect the battery positive (+) to battery positive (+) and the battery negative (−) to battery negative (−).

Jump start only with an energy source with the same voltage as the stalled machine.

Turn off all lights and accessories on the stalled machine. Otherwise, they will operate when the energy source is connected.

NOTICE

When starting from another machine, make sure that the machines do not touch. This could prevent damage to engine bearings and electrical circuits.

Severely discharged maintenance free batteries do not fully recharge from the alternator after jump starting. The batteries must be charged to proper voltage with a battery charger. Many batteries thought to be unusable are still rechargeable.

This machine has a 12 volt starting system. Use only the same voltage for jump starting. Use of a welder or higher voltage damages the electrical system.

Refer to Special Instruction, Battery Test Procedure, SEHS7633, available from your Caterpillar dealer, for complete testing and charging information.

Use of Jump Start Cables

When the auxiliary start receptacles are not available, use the following procedure.

1. Make the initial determination of the machine's failure to crank. The procedure is applicable even if the machine does not have a diagnostic connector.
2. Move the propel control lever of the stalled machine into the NEUTRAL position. Engage the parking brake.
3. Turn the engine start switch on the stalled machine to the OFF position. Turn off all accessories.
4. Move the machine that is being used as a power source so that the jump start cables can reach the stalled machine. **DO NOT ALLOW THE MACHINES TO CONTACT EACH OTHER.**
5. Stop the engine on the machine that is being used as a power source. If you are using an auxiliary power source, turn off the charging system.
6. Check the battery caps for correct placement and for correct tightness. Make these checks on both machines. Make sure that the batteries in the stalled machine are not frozen. Check the batteries for low electrolyte.
7. Connect the positive jump start cable to the positive battery terminal on the stalled machine.

   Do not allow positive cable clamps to contact any metal except for the positive remote jump start terminal.
8. Connect the positive jump start cable to the positive terminal of the boost source.
9. Connect one end of the negative jump start cable to the negative terminal of the electrical source.
10. Make the final connection. Connect the negative cable to the frame of the stalled machine. Make this connection away from the battery, the fuel, the hydraulic lines, or moving parts.
11. Start the engine on the machine that is being used as a power source. If you are using an auxiliary power source, energize the charging system on the auxiliary power source.
12. Wait for a minimum of two minutes while the batteries in the stalled machine partially charge.
13. Attempt to start the stalled engine. Refer to Operation and Maintenance Manual, “Engine Starting”.

14. Immediately after you start the stalled engine, disconnect the jump start cable from the machine that is being used as a power source. Disconnect the negative battery cable first, and then disconnect the positive battery cable.

15. Disconnect the other end of the jump start cable from the stalled machine.

16. When the engine is running and the charging system is in operation, conclude the failure analysis on the starting charging system of the stalled machine, as required.

Maintenance Section

Maintenance Access

i08679098

Access Doors and Covers

SMCS Code: 7251; 7263; 7273-573; 7273; 7273-572

⚠ WARNING

Crush Hazard!

Machine access doors can pinch, trap, or crush personnel when being closed.

Use caution while closing machine access doors. Ensure that all personnel are clear of the machine before closing the access doors.



Illustration 68 g06731519

(1) Latch
(2) Handle

Note: There is one latch (1) on each side of the machine.

Pull latch (2) away from the locking eye and lift on handle (1). Opening the hood will allow access to the engine and major compartments.

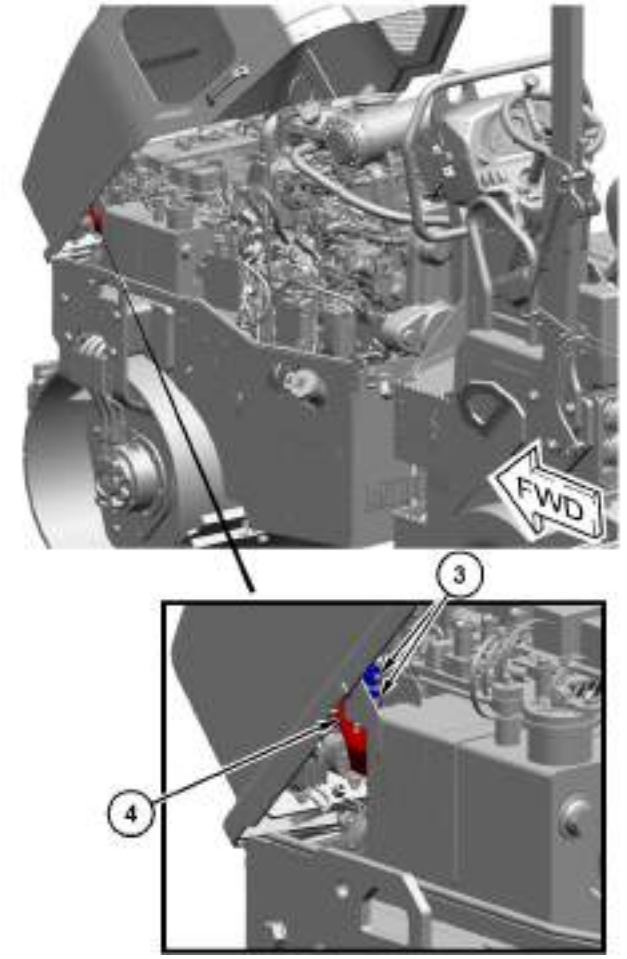


Illustration 69 g06731548

(3) Bolt
(4) Hood lock plate

Loosen bolts (3) and adjust hood lock plate (4) in suitable position. Tighten bolts (3) to hold the hood in open position.

Lubricant Viscosities and Refill Capacities

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Lubricant Viscosities (Fluids Recommendations)

SMCS Code: 1000; 7000; 7581

NOTICE
Not following the recommendations found in this manual can lead to reduced performance and compartment failure.

NOTICE
Never use water alone as a coolant. Water alone is corrosive at engine operating temperatures. In addition, water alone does not provide adequate protection against boiling or freezing.

General Information for Lubricants

When you are operating the machine in temperatures below −20°C (−4°F), refer to Special Publication, SEBU5898, “Cold Weather Recommendations”. This publication is available from your Cat dealer.

For cold-weather applications where transmission oil SAE 0W-20 is recommended, Cat Cold Weather TDTO is recommended.

Refer to the “Lubricant Information” section in the latest revision of the Special Publication, SEBU6250, “Caterpillar Machine Fluids Recommendations” for a list of Cat engine oils and for detailed information. This manual may be found on the Web at Safety.Cat.com.

The footnotes are a key part of the tables. Read ALL footnotes that pertain to the machine compartment in question.

Selecting the Viscosity

In order to select the proper oil for each machine compartment, refer to the “Lubricant Viscosity for Ambient Temperature” table. Use the oil type AND oil viscosity for the specific compartment at the proper ambient temperature.

The proper oil viscosity grade is determined by the minimum ambient temperature (the air in the immediate vicinity of the machine). Measure the temperature when the machine is started and while the machine is operated. In order to determine the proper oil viscosity grade, refer to the “Min” column in the table. This information reflects the coldest ambient temperature condition for starting a cold machine and for operating a cold machine. Refer to the “Max” column in the table for operating the machine at the highest temperature that is anticipated. Unless specified otherwise in the “Lubricant Viscosities for Ambient Temperatures” tables, use the highest oil viscosity that is allowed for the ambient temperature.

Machines that are operated continuously should use oils that have the higher oil viscosity in the final drives and in the differentials. The oils that have the higher oil viscosity will maintain the highest possible oil film thickness. Refer to “General Information for Lubricants” article, “Lubricant Viscosities” tables, and any associated footnotes. Consult your Cat dealer if additional information is needed.

Engine Oil

Cat oils have been developed and tested in order to provide the full performance and life that has been designed and built into Cat engines.

Cat DEO-ULS multigrade and Cat DEO multigrade oils are formulated with the correct amounts of detergents, dispersants, and alkalinity in order to provide superior performance in Cat diesel engines where recommended for use.

Note: SAE 10W-30 is the preferred viscosity grade for the 3116, 3126, C7, C-9, and C9 diesel engines when the ambient temperature is between -18° C (0° F) and 40° C (104° F).

Note: C175 Series diesel engines require the use of **multigrade** SAE 40 oil. For example: SAE 0W-40, SAE 5W-40, SAE 10W-40, or SAE 15W-40. In ambient temperatures of −9.5° C (15° F) or above, SAE 15W-40 is the preferred oil viscosity grade

Table 11

Lubricant Viscosities for Ambient Temperatures						
Compartment or System	Oil Type and Performance Requirements	Oil Viscosities	°C		°F	
			Min	Max	Min	Max
Engine Crankcase	Cat DEO-ULS Cold Weather	SAE 0W-40	−40	40	−40	104
	Cat DEO-ULS SYN Cat DEO SYN	SAE 5W-40	−30	50	−22	122
	Cat DEO-ULS Cat DEO	SAE 10W-30	−18	40	0	104
	Cat DEO-ULS Cat DEO	SAE 15W-40	−9.5	50	15	122

When fuels of sulfur level of 0.1 percent (1000 ppm) or higher are used, Cat DEO-ULS may be used if S·O·S oil analysis program is followed. Base the oil change interval on the oil analysis.

Hydraulic Systems

Refer to the “Lubricant Information” section in the latest revision of the Special Publication, SEBU6250, “Caterpillar Machine Fluids Recommendations” for detailed information. This manual may be found on the Web at Safety.Cat.com.

The following are the preferred oils for use in most Cat machine hydraulic systems:

- Cat HYDO Advanced 10 SAE 10W
- Cat HYDO Advanced 30 SAE 30W
- Cat BIO HYDO Advanced

Cat HYDO Advanced fluids have a 50% increase in the standard oil drain interval for machine hydraulic systems (3000 hours versus 2000 hours) over second and third choice oils when you follow the maintenance interval schedule for oil filter changes and for oil sampling that is stated in the Operation and Maintenance Manual for your particular machine. 6000 hour oil drain intervals are possible when using S·O·S Services oil analysis. Consult your Cat dealer for details. When switching to Cat HYDO Advanced fluids, cross contamination with the previous oil should be kept to less than 10%.

Second choice oils are listed below.

- Cat MTO
- Cat DEO
- Cat DEO-ULS
- Cat TDTO
- Cat TDTO Cold Weather
- Cat TDTO-TMS
- Cat DEO-ULS SYN
- Cat DEO SYN
- Cat DEO-ULS Cold Weather

Table 12

Lubricant Viscosities for Ambient Temperatures						
Compartment or System	**Oil Type and Performance Requirements**	**Oil Viscosities**	**°C**		**°F**	
			Min	**Max**	**Min**	**Max**
Hydraulic System	**Cat HYDO Advanced 10** Cat TDTO	SAE 10W	−20	40	−4	104
	Cat HYDO Advanced 30 Cat TDTO	SAE 30	0	50	32	122
	Cat BIO HYDO Advanced	“ISO 46” Multi-Grade	−30	45	−22	113
	Cat MTO Cat DEO-ULS Cat DEO	SAE10W-30	−20	40	−4	104
	Cat DEO-ULS Cat DEO	SAE15W-40	−15	50	5	122
	Cat TDTO-TMS	Multi-Grade	−15	50	5	122
	Cat DEO-ULS SYN Cat DEO SYN	SAE 5W-40	−25	40	−13	104
	Cat DEO-ULS Cold Weather	SAE0W-40	−40	40	−40	104
	Cat TDTO Cold Weather	SAE 0W-20	−40	40	−40	104

Special Lubricants

Grease

In order to use a non-Cat grease, the supplier must certify that the lubricant is compatible with Cat grease.

Each pin joint should be flushed with the new grease. Ensure that all old grease is removed. Failure to meet this requirement may lead to failure of a pin joint.

Table 13

Recommended Grease						
Eccentric lubrication points	**GreaseType**	**NLGI Grade**	**°C**		**°F**	
			Min	**Max**	**Min**	**Max**
Eccentric lubrication points	**Cat Ultra 5Moly**	NLGI Grade 2	−30	50	−22	122
		NLGI Grade 1	−35	40	−31	104
		NLGI Grade 0	−40	35	−40	95
External lubrication points	**Cat Arctic Platinum**	NLGI Grade 0	−50	20	−58	68
	Cat Advanced 3Moly	NLGI Grade 2	−20	40	−4	104
	Cat Desert Gold	NLGI Grade 2	−20	60	−4	140

Diesel Fuel Recommendations

Diesel fuel must meet “Caterpillar Specification for Distillate Fuel” and the latest versions of “ASTM D975” or “EN 590” in order to ensure optimum engine performance. Refer to Special Publication, SEBU6250, “Caterpillar Machine Fluids Recommendations” for the latest fuel information and for Cat fuel specification. This manual may be found on the Web at Safety.Cat.com.

The preferred fuels are distillate fuels. These fuels are commonly called diesel fuel, furnace oil, gas oil, or kerosene. These fuels must meet the “Caterpillar Specification for Distillate Diesel Fuel for Off-Highway Diesel Engines”. Diesel Fuels that meet the Caterpillar specification will help provide maximum engine service life and performance.

Misfueling with fuels of high sulfur level can have the following negative effects:

- Reduce engine efficiency and durability
- Increase the wear
- Increase the corrosion
- Increase the deposits
- Lower fuel economy
- Shorten the time period between oil drain intervals (more frequent oil drain intervals)
- Increase overall operating costs
- Negatively impact engine emissions

Failures that result from the use of improper fuels are not Caterpillar factory defects. Therefore the cost of repairs would not be covered by a Caterpillar warranty.

Caterpillar does not require the use of ULSD in off road and machine applications that are not Tier 4/ Stage IIIB certified engines. ULSD is not required in engines that are not equipped with after treatment devices.

Follow operating instructions and fuel tank inlet labels, if available, in order to ensure that the correct fuels are used.

Refer to Special Publication, SEBU6250, “Caterpillar Machine Fluids Recommendations” for more details about fuels and lubricants. This manual may be found on the Web at Safety.Cat.com.

Fuel Additives

Cat Diesel Fuel Conditioner and Cat Fuel System Cleaner are available for use when needed. These products are applicable to diesel and biodiesel fuels. Consult your Cat dealer for availability.

Biodiesel

Biodiesel is a fuel that can be made from various renewable resources that include vegetable oils, animal fat, and waste cooking oil. Soybean oil and rapeseed oil are the primary vegetable oil sources. In order to use any of these oils or fats as fuel, the oils or fats are chemically processed (esterified). The water and contaminants are removed.

U.S. distillate diesel fuel specification “ASTM D975-09a” includes up to B5 (5 percent) biodiesel. Currently, any diesel fuel in the U.S. may contain up to B5 biodiesel fuel.

European distillate diesel fuel specification “EN 590” includes up to B5 (5 percent) and in some regions up to B7 (7 percent) biodiesel. Any diesel fuel in Europe may contain up to B5 or in some regions up to B7 biodiesel fuel.

When biodiesel fuel is used, certain guidelines must be followed. Biodiesel fuel can influence the engine oil, aftertreatment devices, non-metallic, fuel system components, and others. Biodiesel fuel has limited storage life and has limited oxidation stability. Follow the guidelines and requirements for engines that are seasonally operated and for standby power generation engines.

In order to reduce the risks associated with the use of biodiesel, the final biodiesel blend and the biodiesel fuel used must meet specific blending requirements.

All the guidelines and requirements are provided in the latest revision of Special Publication, SEBU6250, “Caterpillar Machine Fluids Recommendations”. This manual may be found on the Web at Safety.Cat.com.

Coolant Information

The information provided in this “Coolant Recommendation” section should be used with the “Lubricants Information” provided in the latest revision of Special Publication, SEBU6250, “Caterpillar Machine Fluids Recommendations”. This manual may be found on the Web at Safety.Cat.com.

The following two types of coolants may be used in Cat diesel engines:

Preferred – Cat ELC (Extended Life Coolant)

Acceptable – Cat DEAC (Diesel Engine Antifreeze/ Coolant)

Standard Factory Fill Fluids

Table 14

Standard Factory Fill Fluids[1]					
Compartment or System	**Oil Viscosities**	**°C**		**°F**	
		Min	**Max**	**Min**	**Max**
Engine Crankcase	Cat 10-30 ULS	−9.5	50	15	122
Hydraulic Systems	Cat HYDO Advanced 10	-20	40	-4	104
Coolant	Cat ELC 50% / 50%	-36		-33	

(1) The machine is delivered from the factory with the designated fluids.

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Capacities (Refill)

SMCS Code: 1000; 6320; 7000; 7560

Table 15

REFILL CAPACITIES Approximate			
Compartment or System	**Liters**	**US Gallon**	**Imperial Gallon**
Fuel Tank	54	14.3	11.9
Engine Oil	6.3	1.7	1.4
Hydraulic Tank Oil	21	5.5	4.6
Water Spray Tank CB2.5, CB2.7, CB2.9	198	52	44
Water Spray Tank CC2.7	170	44.9	37.4
Additive Spray Tank CC2.7	24	6.3	5.3
Coolant	8.4	2.2	1.8

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S·O·S Information

SMCS Code: 1348; 3080; 4070; 4250; 4300; 5050; 7542

S·O·S Services is a highly recommended process for Cat customers to use in order to minimize owning and operating cost. Customers provide oil samples, coolant samples, and other machine information. The dealer uses the data in order to provide the customer with recommendations for management of the equipment. In addition, S·O·S Services can help determine the cause of an existing product problem.

Refer to Special Publication, SEBU6250, “Caterpillar Machine Fluid Recommendations” for detailed information concerning S·O·S Services.

The effectiveness of S·O·S Services is dependent on timely submission of the sample to the laboratory at recommended intervals.

Refer to the Operation and Maintenance Manual, “Maintenance Interval Schedule” for a specific sampling location and a service hour maintenance interval.

Consult your Cat dealer for complete information and assistance in establishing an S·O·S program for your equipment.

Maintenance Support

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System Pressure Release

SMCS Code: 1250-553-PX; 1300-553-PX; 1350-553-PX; 3000-553-PX; 4250-553-PX; 4300-553-PX; 5050-553-PX; 7540-553-PX

WARNING

Personal injury or death can result from sudden machine movement.

Sudden movement of the machine can cause injury to persons on or near the machine.

To prevent injury or death, make sure that the area around the machine is clear of personnel and obstructions before operating the machine.

Coolant System

WARNING

Pressurized system: Hot coolant can cause serious burn. To open cap, stop engine, wait until radiator is cool. Then loosen cap slowly to relieve the pressure.

To relieve the pressure from the coolant system, turn off the machine. Allow the cooling system pressure cap to cool. Remove the cooling system pressure cap slowly in order to relieve pressure.

Engine Oil System

To relieve the pressure from the engine oil system, turn off the engine.

Fuel System

To relieve the pressure from the fuel system, turn off the engine.

High Pressure Fuel Lines

WARNING

Contact with high pressure fuel may cause fluid penetration and burn hazards. High pressure fuel spray may cause a fire hazard. Failure to follow these inspection, maintenance and service instructions may cause personal injury or death.

The high-pressure fuel lines are the fuel lines that are between the high-pressure fuel pump and the high-pressure fuel manifold and the fuel lines that are between the fuel manifold and cylinder head. These fuel lines are different from fuel lines on other fuel systems because of the following differences:

- The high-pressure fuel lines are constantly charged with high pressure.
- The internal pressures of the high-pressure fuel lines are higher than other types of fuel system.

Before any service or repair is performed on the engine fuel lines, perform the following tasks:

1. Turn off the engine.

2. Wait for 10 minutes.

Note: Fuel pressure can be monitored by Caterpillar Electronic Technician (ET).

Do not loosen the high-pressure fuel lines in order to purge trapped air from the fuel system.

Hydraulic System

WARNING

Personal injury can result from hydraulic oil pressure and hot oil.

Hydraulic oil pressure can remain in the hydraulic system after the engine has been stopped. Serious injury can be caused if this pressure is not released before any service is done on the hydraulic system.

Make sure all of the attachments have been lowered, oil is cool before removing any components or lines. Remove the oil filler cap only when the engine is stopped, and the filler cap is cool enough to touch with your bare hand.

1. Shut off the engine.

2. Apply the parking brake.

3. Slowly loosen the filler cap in order to release the pressure in the hydraulic tank.

4. Tighten the filler cap.

5. The pressure in the hydraulic system has been released. Lines and components can be removed.

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Welding on Machines and Engines with Electronic Controls

SMCS Code: 1000; 7000

Do not weld on any protective structure. If it is necessary to repair a protective structure, contact your Cat dealer.

Proper welding procedures are necessary to avoid damage to the electronic controls and to the bearings. When possible, remove the component that must be welded from the machine or the engine and then weld the component. If you must weld near an electronic control on the machine or the engine, temporarily remove the electronic control to prevent heat related damage. The following steps should be followed to weld on a machine or an engine with electronic controls.

1. Turn off the engine. Place the engine start switch in the OFF position.

2. If equipped, turn the battery disconnect switch to the OFF position. If there is no battery disconnect switch, remove the negative battery cable at the battery.

NOTICE
Do NOT use electrical components (ECM or sensors) or electronic component grounding points for grounding the welder.

3. Clamp the ground cable from the welder to the component that will be welded. Place the clamp as close as possible to the weld. Make sure that the electrical path from the ground cable to the component does not go through any bearing. Use this procedure to reduce the possibility of damage to the following components:

- Bearings of the drive train
- Hydraulic components
- Electrical components
- Other components of the machine

4. Protect any wiring harnesses and components from the debris and the spatter which is created from welding.

5. Use standard welding procedures to weld the materials together.

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Prepare the Machine for Maintenance

SMCS Code: 1000; 7000

1. Move the machine to a dry, level, solid surface that is free of any debris.

Note: The surface must be solid enough to support the weight of the machine and any tooling that is used to support the machine.

2. Put the machine in park. Refer to Operation and Maintenance Manual, “Operator Controls” for more information.

3. Ensure that the pressure is released from any closed system that will be opened during the maintenance procedure. Refer to Operation and Maintenance Manual, “System Pressure Release” for more information.

Maintenance with the Engine Running

For maintenance that requires the engine to be running, perform the following:

1. Run the engine at an idle.

Maintenance without the Engine Running

For maintenance that does not require the engine to be running, perform the following:

1. Move the engine start switch to the OFF position. Refer to Operation and Maintenance Manual, “Operator Controls” for more information.

Maintenance with Electrical System Disabled

For maintenance that requires the electrical system to be disabled, perform the following:

1. Move the engine start switch to the OFF position. Refer to Operation and Maintenance Manual, “Operator Controls” for more information.

2. Move the battery disconnect switch to the OFF position. Refer to Operation and Maintenance Manual, “Battery Disconnect Switch” for the proper procedure.

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Maintenance Interval Schedule

SMCS Code: 1000; 7000

Ensure that all safety information, warnings, and instructions are read and understood before any operation or any maintenance procedures are performed.

The user is responsible for the performance of maintenance. All adjustments, the use of proper lubricants, fluids, filters, and the replacement of components due to normal wear and aging are included. Failure to adhere to proper maintenance intervals and procedures may result in diminished performance of the product and/or accelerated wear of components.

Products that operate in severe operating conditions or that experience abnormally high fuel consumption may require more frequent maintenance. Refer to the maintenance procedure for any other exceptions that may change the maintenance intervals.

Note: Before each consecutive interval is performed, all maintenance from the previous interval shall be performed.

Note: The aftertreatment system can be expected to function properly for the useful life of the engine (emissions durability period), as defined by regulation. All prescribed maintenance requirements shall be followed.

The following guidelines should be followed if the service hours are not met:

- Items listed between 10 and 100 service hours should be performed at least every 3 months.
- Items listed between 250 and 500 service hours should be performed at least every 6 months.
- Items listed between 1000 service hours and 2500 service hours should be performed at least every year.

When Required

“ Additive Spray System - Drain“ 77
“ Additive Spray System - Prime“ 78
“ Battery - Recycle“ 81
“ Battery or Battery Cable - Inspect/Replace“. 82
“ Engine - Clean“ 100
“ Engine Air Filter Primary Element - Clean/ Replace“ 101
“ Engine Air Filter Secondary Element - Replace“ 103
“ Film (Product Identification) - Clean“ 109
“ Fuel System - Prime“ 111
“ Fuses - Replace“ 118
“ Oil Filter - Inspect“ 131
“ Radiator Core - Clean“ 132
“ Water and Additive Spray Nozzles - Clean“ 136
“ Water Spray System - Drain“ 137
“ Wheel Nuts - Tighten“ 140

Every 10 Service Hours or Daily

“ Backup Alarm - Test“ 79
“ Cooling System Coolant Level - Check“. 90
“ Drum Scrapers - Inspect/Adjust/Replace“ 98
“ Engine Oil Level - Check“ 105
“ Hydraulic System Oil Level - Check“ 126
“ Neutral Start Switch - Test“ 131
“ Operator Presence Switch - Test“. 132
“ Seat Belt - Inspect“ 133
“ Tire Scraper - Inspect/Adjust/Replace“ 135
“ Water Spray System Filter - Clean“ 138

Initial 50 Service Hours

“ Wheel Nuts - Tighten“ 140

Every 50 Service Hours

“ Fuel Tank Water and Sediment - Drain“ 118
“ Tire Inflation - Check“ 134

Every 100 Service Hours

“ Water Tank Strainer - Clean and Inspect“ 139

Every 250 Service Hours

“ Belts - Inspect/Adjust/Replace“. 83
“ Cooling System Coolant Sample (Level 1) - Obtain“ 92
“ Engine Oil Sample - Obtain“ 106